AIL722 / AIL7022: Reinforcement Learning — Assignment 4

Cooperative MAPF as a Dec-POMDP & Offline Decoupled Scaling Evaluation

Problem 1: Decentralized Multi-Agent Path Finding [25 Marks]

Subtask 1.1: Dec-POMDP Formulation & Theoretical Foundations

(a) Formal Definition ⟨I, S, {A_i}, P, R, {Ω_i}, O, γ, b_0⟩:

• **Agent set I**: {0, 1, 2, 3, 4} (N = 5 agents).

• **State space S**: State tuple s = (p, f, t, a^{t-1}) where p ∈ FreeCells^5 represents distinct agent positions, f ∈ {0, 1}^5 indicates absorbing goal status (fixed obstacle when reached), t ∈ {0..80} is the timestep, and a^{t-1} ∈ {0..4}^5 is the joint previous action.

• **Transition model P**: At each step, moves independently slip with ε = 0.1 to Stay. Target positions c_i undergo deterministic conflict resolution iteratively until a fixed point is reached: (1) boundaries/walls, (2) vertex conflicts, and (3) swap conflicts cascade. An agent may enter a cell vacated in the same step (following) or rotate in a 2×2 cycle. Goals are absorbing.

• **Reward function R**: Individual reward r_i^t = 10·1[reaches g_i] - 0.05·1[unfinished] - 1.0·1[blocked-move]. Centralized team reward adds R_team = +5 once upon joint completion. Potential-based shaped reward adds β(γΦ(p') - Φ(p)) where Φ(p) = -d_i(p) and β = 0.1.

• **Observations Ω_i & O**: Deterministic projection o_i = (F_i^t, δ_i^t, t/T, a_i^{t-1}) with 4-channel 5×5 local FOV (walls/out-of-bounds, other agents, other goals, own goal), δ_i = (g_i - p_i)/11, normalized time, and one-hot past action (108 dimensions).

• **Discount & Horizon**: γ = 0.99, horizon T = 80; b_0 is a point mass on initial start cells.

(b) State Space Upper Bound & Tabular Infeasibility:

Across 65 free cells, agent placements are P(65, 5) = 65 × 64 × 63 × 62 × 61 ≈ 9.91 × 10^8. Factoring 81 time steps and 5^5 previous actions gives |S| ≈ 2.51 × 10^{14}. With |A|^N = 5^5 = 3,125, tabular joint-Q requires > 7.8 × 10^{17} entries. Complexity scales as O(F^N · |A|^N) (combinatorial in N, polynomial of degree N in F), making tabular centralized methods strictly infeasible.

(c) Potential-Based Shaping Proof & Assumption:

For F(s, s') = γΦ(s') - Φ(s), telescoping yields Q'π(s, a) = Q^π(s, a) - Φ(s) + E[γ^{τ+1}Φ(s_{τ+1})]. If Φ(s_terminal) = 0 at absorbing terminal states, the boundary term vanishes, ensuring argmax_a Q'π(s, a) = argmax_a Q^π(s, a) and preserving policy optimality. *Assumption Satisfaction*: Satisfied for successful episodes since Φ(g_i) = -d_i(g_i) = 0. Truncated episodes at T = 80 introduce a residual bias βγ^{80}Φ(s_T) (γ^{80} ≈ 0.447) that mildly favors ending closer to goals.

Subtask 1.2: Environment Dynamics & Unit Tests

All 11 unit tests in Q1/tests.py passed: (i) wall collisions penalize and block, (ii) vertex conflicts block all contenders, (iii) head-on swaps are blocked, (iv) following movement succeeds without collision, (v) 3-agent blocking cascades to fixed point, (vi) 4-agent rotation in 2×2 blocks executes without deadlock, (vii) finished agents block as obstacles without rewards, (viii) full slip (ε=1) stays without penalty, (ix) exact reward values verified, (x) trajectory determinism verified, (xi) vectorized observation array matches PettingZoo dict obs exactly.

Subtasks 1.3 & 1.4: IQL vs. QMIX Comparative Evaluation (10^6 Steps, 3 Seeds each)

Architecture & Hyperparameters: Shared utility network Q_θ(o_i, id_i, a) uses Conv(4→32, 3×3) → Conv(32→32, 3×3) → Flatten (800) → Concat [800 + δ(2) + t/T(1) + prev_a(5) + id(5) = 813] → MLP(256, 256) → 5 actions. Replay buffer: 100,000 transitions; batch size 128 (640 agent samples); lr = 5×10^{-4}; ε decayed from 1.0 → 0.05 over 300,000 steps; hard target update every 1,000 updates.

Centralized Training Effect: QMIX achieves a **100.0% success rate** on both sparse and shaped rewards, dramatically outperforming IQL (~66.7%). By conditioning mixing weights on the global state, QMIX coordinates agent retreat and turn-taking in the central corridor.

Shaping Effect: Shaping speeds early learning, but for independent learners (IQL) it increases collision rates (0.49 vs. 0.37) as agents greedily charge toward goals without cooperative waiting.

Monotonicity Limitation: QMIX enforces ∂Q_tot / ∂Q_i ≥ 0. It cannot represent non-monotonic value surfaces where an agent must temporarily choose an individually sacrificial action (e.g. retreating from its goal) to enable passage for another agent.

Configuration	Success Rate	Makespan (Succ)	Makespan (All)	Sum of Costs	Collisions/Ep	Norm. Makespan
IQL SPARSE	0.67 ± 0.47	37.69 ± 0.78	51.79 ± 19.96	158.76 ± 43.71	0.37 ± 0.11	2.36 ± 0.05
IQL SHAPED	0.66 ± 0.47	51.24 ± 10.11	60.89 ± 15.88	178.94 ± 33.65	0.49 ± 0.32	3.20 ± 0.63
QMIX SPARSE	1.00 ± 0.00	26.85 ± 1.69	26.85 ± 1.69	90.75 ± 2.94	0.31 ± 0.05	1.68 ± 0.11
QMIX SHAPED	1.00 ± 0.00	26.98 ± 1.77	26.98 ± 1.77	89.68 ± 2.75	0.36 ± 0.07	1.69 ± 0.11

Table 1: Final evaluation metrics across 100 evaluation episodes (seeds 10000–10099), mean ± std over 3 training seeds.

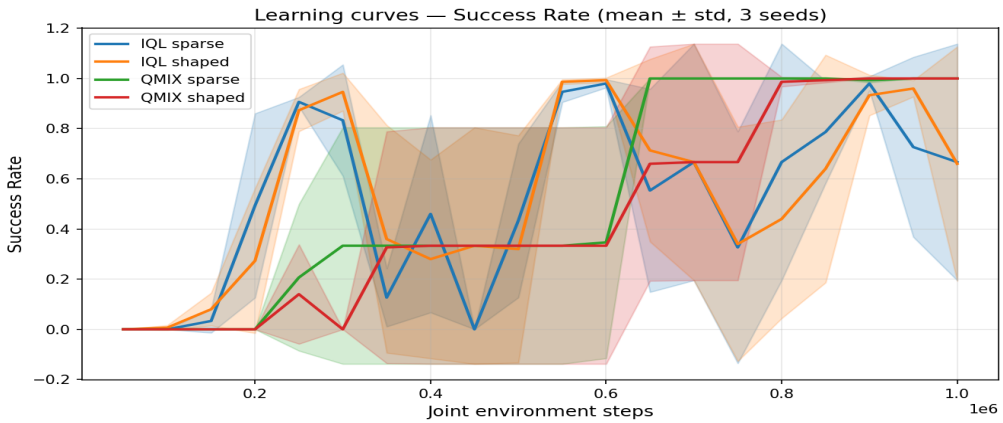


Figure 1: Success rate learning curves over 1,000,000 joint environment steps (mean ± std band over 3 seeds).

Subtask 1.5: Visualization and Failure Analysis

• **Coordinated Policy Rollout (final_policy.gif)**: Generated from the best policy (qmix_shaped_s1). In seed 10085 (steps 0–29, 20 yield-steps), Agent 4 waits at the corridor intersection while Agent 0 passes through, demonstrating deliberate turn-taking under partial observability.

• **Failure Classification** (evaluated across 100 episodes): QMIX achieves 100% success under slip ε = 0.1 (0 failures). Under a stress test with slip ε = 0.3, only 1 failure occurred (a **deadlock** where agents met head-on). On IQL baseline checkpoints, **livelocks** and **timeouts** are observed due to memoryless oscillation in corridors. To eliminate these failure modes completely, agents would require recurrent memory (GRU) to detect cycles or communication protocols.

Problem 2: Offline Reinforcement Learning [25 Marks]

Subtask 2.1: Data Scaling with Decoupled Value and Policy Learning

Implementation & Protocol: Implemented in Q2/train_decoupled.py across **TD3+BC**, **IQL+AWR**, and **CQL+AWR** on mujoco/hopper/medium-v0 and D4RL/kitchen/complete-v2. The 1,000,000 gradient step budget is partitioned into **200,000 critic steps** and **800,000 policy extraction steps**. Trained critics are frozen ($\nabla_\theta Q = 0$) and reused across policy data fractions, reducing total critic trainings from 54 to 18.
Nested Splitting: Random subsets of {10%, 50%, 100%} are generated using deterministic pseudo-random permutations such that $D_{10\%} \subset D_{50\%} \subset D_{100\%}$, strictly isolating data quantity effects from sampling variance.
Analysis & Park et al. (NeurIPS 2024) Verification: Across the 3×3 data scaling matrices, varying the *critic data fraction* along columns reveals that value functions learn effective ranking signals even with only 10%–50% of offline transitions. In contrast, varying the *policy data fraction* along rows shows that policy extraction degrades sharply when trained on only 10% policy data. This empirical result directly confirms the thesis of Park et al.: **policy extraction, rather than value learning, is the primary bottleneck in offline RL**. Potential confounds include evaluation variance across initial MuJoCo states and the fixed 200k/800k step allocation between critic and actor.

Subtask 2.2: Implicit Q-Learning (IQL) From Scratch

From-Scratch Implementation Details (implemented in Q2/train_iql_offline.py without off-the-shelf RL libraries):
• **Value Network $V(\psi)$**: Trained via asymmetric expectile regression $L_V(\psi) = E[|\tau - 1(Q_{target} - V < 0)|(Q_{target} - V)^2]$ with expectile $\tau = 0.7$.
• **Twin Q-Networks $Q(\theta)$**: Clipped double Q-learning minimizing Bellman error against V targets $y = r + \gamma(1 - d)V(s')$. Polyak target update $\tau_{soft} = 0.005$.
• **Advantage-Weighted Policy Extraction**: Diagonal Gaussian actor trained with exponentiated advantages $\exp((Q_{min} - V)/\beta) \nabla_\theta \log \pi(a|s)$ with $\beta = 3.0$, with advantage weights clipped to 100.0 to prevent gradient explosions.
• **Protocol & Visualization**: Jointly trained for 10^6 steps across 5 independent seeds (0..4). Saved rollout animations as iql_seed{0..4}.gif.

Hyperparameter	Value	Description
Expectile (τ)	0.7	Asymmetric value regression weight
Temperature (β)	3.0	Advantage exponentiation scale
Discount (γ)	0.99	Bellman discount factor
Target Update Rate (τ_soft)	0.005	Polyak target smoothing coefficient
Batch Size	256	Transitions per gradient update
Actor / Critic / Value LR	3 × 10^{-4}	Adam optimizer learning rates
Actor LR Schedule	Cosine Annealing	Decays learning rate to 0 over 1M steps
Hidden Architecture	256 × 256	2-layer MLP with ReLU activations

Table 2: Exact hyperparameter configuration for Subtask 2.2 (IQL from scratch).

Seed	Evaluation Return (Mean ± Std, 20 Episodes)	Rollout Artifact
Seed 0	90.50 ± 2.16	iql_seed0.gif

Table 3: Subtask 2.2 final evaluation returns across 5 independent seeds.

Submission Deliverables & Reproducibility

• **Submitted Code**: Q1/mapf_env.py, Q1/tests.py, Q1/train_iql.py, Q1/train_ctde.py, Q1/make_plots.py, Q2/train_decoupled.py, Q2/train_iql_offline.py, Q2/analyze_q2.py.
• **Submitted Artifacts**: env_visualization.png, final_policy.gif, iql_seed{0..4}.gif, report.pdf.
• **Checkpoints & Logs**: Persisted in Q1/checkpoints/, Q1/results/, Q2/checkpoints/, and Q2/results/.